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Behavioral and phylogenetic correlates of limb length proportions in extant apes and monkeys: Implications for interpreting hominin fossils

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ABSTRACT

The body proportions of extant animals help inform inferences about the behaviors of their extinct relatives, but relationships between body proportions, behavior, and phylogeny in extant primates remain unclear. Advances in behavioral data, molecular phylogenies, and multivariate analytical tools make it an opportune time to perform comprehensive comparative analyses of primate traditional limb length proportions (e.g., intermembral, humerofemoral, brachial, and crural indices), body size-adjusted long bone proportions, and principal components. In this study we used a mix of newly-collected and published data to investigate whether and how the limb length proportions of a diverse sample of primates, including monkeys, apes, and modern humans, are influenced by behavior and phylogeny. We reconfirm that the intermembral index, followed by the first principal component of traditional limb length proportions, is the single most effective variable distinguishing hominoids and other anthropoids. Combined limb length proportions and positional behaviors are strongly correlated in extant anthropoid groups, but phylogeny is a better predictor of limb length proportion variation than of behavior. We confirm convergences between members of the Atelidae and extant apes (especially *Pan*), members of the Hylobatidae and *Pongo*, and a potential divergence of *Presbytis* limb proportions from some other cercopithecoids, which correlate with adaptations for forelimb-dominated behaviors in some colobines. Collectively, these results substantiate hypotheses indicating that extinct hominins and other hominoid taxa can be distinguished by analyzing combinations of their limb length proportions at different taxonomic levels. From these results, we hypothesize that fossil skeletons characterized by notably disparate limb length proportions are unlikely to have exhibited similar behavioral patterns.

1. Introduction

Limb proportions have been used to infer important aspects of early positional hominin behavior (i.e., locomotion and posture; Prost, 1965), but gaps in our knowledge presently hinder their effective use. Limb length proportions (the relative lengths of the limbs and their segments: e.g., intermembral, humerofemoral, brachial, and crural ratios) have been crucial in primate comparative studies since Tyson's (2018) recognition that differences in body proportions among apes, monkeys, and modern humans are linked to their behavior. Mollison (1910) and Schultz (1930, 1933, 1937, 1973) published detailed morphological data linking the diversity of primate limb length proportions to various aspects of their biology. Napier (1967) and Napier and Napier (1967, 1985) defined categories of extant primate locomotion and linked them to adaptive trends in body size and limb length proportions, with a focus on the intermembral index. Subsequently, Napier and Napier (1967, 1985; see also Napier and Walker (1967) became commonly cited to support hypotheses about locomotor mode in extinct hominins (Asfaw et al., 1999; Hunt et al., 1996; Jungers, 1985; Lordkipanidze et al., 2007; Lovejoy et al., 2009; Richmond et al., 2002; Wright et al., 2015). Some criticized the locomotor categories presented by Napier and colleagues as being broad and/or inaccurate (e.g., Cartmill, 1974; Fleagle, 1988; Mittermeier and Fleagle, 1976), but most researchers still support a link between predominant locomotor mode and limb length proportions, although that link has yet to be quantified in a comprehensive analysis of extant primates (e.g., Biegert and Maurer, 1972; Fleagle, 1988, 2013;

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Fleagle and Mittermeier, 1980; Jungers, 1985).

A lack of standardization in recording methods and behavioral categories has been an obstacle to collecting comprehensive behavioral data about wild primates, and thus an obstacle to quantifying the link to limb proportions. Hunt et al. (1996), using both their own and published observations, proposed using 32 positional modes subdivided into 74 locomotor and 52 postural sub-modes, to provide standardized categories of primate positional and locomotor behavior. In a more recent publication, Hunt (2016) provided the frequencies with which 33 extant hominoid and cercopithecoid taxa use the standardized behaviors in their positional repertoires, but as yet there has been no attempt to examine the relationship between limb length proportions and Hunt's classification in a diverse range of extant primates.

Researchers have investigated the effects of ontogeny, sex, phylogeny, environment, and scaling on limb length and cross-sectional proportions, locomotor behaviors, and functional morphology of the bones and joints of extant primates (Carlson and Pickering, 2004; Druelle and Berillon, 2013; Gordon et al., 2020; Hurov, 1987; Jungers et al., 1998; O'Neill and Dobson, 2008; Patel et al., 2013; Ramírez-Llorens et al., 2008; Ruff, 2002; Ruff et al., 2018, 2019, 2022; Ruff and Runestad, 1992; Takahashi, 1990; Turnquist et al., 1999; Vereecke et al., 2011). However, studies linking limb length proportions to the behaviors of primates in the wild are often restricted to comparisons among closely-related or sympatric species, or the methods used are not quantitative, or the comparisons are limited to one region of the postcranial skeleton (Strasser, 1992; Jungers, 1979; Nakatsukasa, 1996; Anapol et al., 2005; Su and Jablonski, 2009; Byron et al., 2017; Kimura, 2021; see also Polk, 2004). For example, although Rose (1996) and Wright et al. (2015; see also Wright, 2007) describe the limb proportions of extant platyrrhines and Miocene catarrhines, and capuchins, respectively, they avoid exploring quantitative links between morphology and behavior. Ford and Hobbs (1996) distinguish dietary, locomotor, and skeletal traits including limb bone lengths among *Cebus* species, but do not quantitatively support the observed differences, and the Rein et al. (2015) exploration of shape covariation to infer forelimb suspensory behavior in extinct primates was restricted to the ulna.

Simple mechanical principles described in Napier and Napier (1967) predict that relatively longer fore- or hindlimbs permit, for instance, relatively longer reaches or strides and greater acceleration during swinging or leaping and bipedalism, respectively (but posture, limb excursion, and body mass are other important, mediating factors; Preuschoft, 2002; Polk, 2004). Because of their clear links with limb dominance in primates, the intermembral index and its close approximate, the humerofemoral index, are frequently used to study locomotor differences, and the brachial, and crural indices are also informative about primate behavioral repertoires involving suspension and bipedalism due to similar rules (Anapol et al., 2005; Byron et al., 2017; Fleagle and Meldrum, 1988). However each index captures only part of the overall picture and, on its own, fails to capture body mass information and other limb segment information influencing a group's positional repertoire, thus they have been criticized as sole predictors of positional behaviors (Jungers, 1979, 1984, 1985; Tallman, 2013). Oxnard et al. (1981) combined multiple postcranial indices in discriminant analyses to reveal previously unrecognized group structures among prosimians, and Young (2003) used multivariate cladistic, principal component analyses, and cluster analyses on a large set of postcranial variables to demonstrate the same in the three anthropoid superfamilies. However, none of the traditional limb length proportions directly account for body size, which is a major selective factor affecting behavior, limb scaling, and function that is likely to affect results. Analyses of relative limb lengths that account for body size using proxy measurements (e.g., geometric means) could help improve efforts to use limb length data to retrodict the locomotor and postural behaviors of extinct primates including early hominins.

Here, we use a large, phylogenetically, behaviorally, and morphologically diverse anthropoid dataset to investigate how behavior and phylogeny influence limb proportions. We used a multivariate approach to understand overall appendicular shape by analyzing combinations of traditional limb bone length indices and size-adjusted ratios as shape variables. We build upon previous studies (e.g., Napier and Napier, 1967; Napier and Walker, 1967; Fleagle, 1988; see also Jungers, 1985; Tallman, 2013; Almécija et al., 2015; Hunt, 2016) by utilizing standardized locomotor categories based on published field observations to quantify multivariate correlations between overall limb length proportions and behavioral repertoires in a comprehensive comparative sample that includes multiple genera of all of the extant anthropoid superfamilies. We address three main research questions:

1) How well can extant anthropoid taxa be differentiated using limb bone length proportions alone, and which limb length proportions best separate them? If individual, or combined, limb proportions reliably separate large, phylogenetically diverse, samples of anthropoid taxa, then it is parsimonious to expect that the same is true of their fossil close relatives at equivalent taxonomic levels. We will test the hypothesis that limb length proportions differentiate extant anthropoid taxa using individual ratios and major axes of variation, with a focus on genera. We predict that major axes of multivariate variation differentiate taxa within our sample at least as well as individual limb proportions. Consistent with existing literature (e.g., Jungers, 1984, 1985), we also predict that of all individual ratios, intermembral and humerofemoral indices are most effective for distinguishing taxa.
2) To what extent do limb length proportions correlate with behavioral repertoires in extant anthropoids? Phenotypic convergence (e.g., *Hylobates*-like postcranial dimensions and forelimb-dominated locomotion in atelids; see Larson, 1998; Young, 2003), and unexpected behavior based on phenotype and phylogeny (e.g., modern human-like limb proportions and rare suspensory behaviors in colobines; Gebo and Chapman, 1995; Hunt, 2016; Byron et al., 2017) indicate a complex relationship between morphology and behavior. These relationships have yet to be quantified across extant anthropoids, which calls into question inferences about their fossil relatives. We will assess associations between limb length proportions and positional behaviors in extant anthropoids. Based on the extensive published literature, we hypothesize that taxa using a relatively greater frequency of forelimb-dominated behaviors have relatively elongated forelimb elements, and taxa using a relatively greater frequency of hindlimb-dominated behaviors have relatively elongated hindlimb elements.
3) How much of the observed variations in limb length proportions and positional behaviors are explained by phylogeny? Limb lengths are recognized to be under strong genetic control (Young et al., 2010; Rolian, 2014), but the extent to which phylogeny explains differences in limb length proportions and positional behaviors in extant hominoids and other anthropoids remains incompletely explored (but see O'Neill and Dobson, 2008 regarding phylogenetic signals in limb lengths of non-hominoids; Ruff et al., 2019 regarding behavioral correlates of articular and cross-sectional properties in cercopithecoid monkeys). This information is crucial for accurately interpreting fossil limb proportions in terms of ancestry, behavioral specializations, locomotor transitions, or convergence. We will assess hypotheses regarding the relationships among limb proportions, locomotor behaviors, and phylogenetic structure, and determine the strength of the resulting signals.

Our aim is to better understand the links between positional behavior, appendicular morphology, and phylogeny in extant anthropoids to improve future predictions regarding extinct hominins.

2. Materials and methods

2.1. Sample

Anthropoid taxa representing the major locomotor groups described by Napier and Napier (1967) were selected based on available field observations of positional behavior. The sample ($n = 1825$) includes a diverse selection of non-human anthropoids ($n = 555$) with approximately equal numbers of adult males and females for each taxon, as well as a large, regionally heterogeneous, and phenotypically diverse sample of modern humans ($n = 1270$, comprising 789 males and 481 females) spanning from 5500 years ago to the present (data summary in Table 1; complete dataset in Supplementary Online Material [SOM] Table S1). The non-human anthropoid sample comprises five hominoid ($n = 312$), nine cercopithecoid ($n = 153$), and 12 ceboid ($n = 90$) genera. The majority of the modern human data were obtained from the Goldman Osteometric Dataset (Auerbach and Ruff, 2004, 2006). Additional information is available on their website (https://web.utk.edu/~auerbach/GOLD.htm; Auerbach and Ruff, 2004, 2006). The remaining modern human data was collected by the first author from the Robert J. Terry Anatomical Collection of the Smithsonian National Museum of Natural History. Descriptive statistics are provided at the genus level in SOM Table S2.

Table 1
Anthropoid sample included in the current study.

Superfamily	Genus	Species	Males	Females	Unknown	*n*
Hominoidea	*Homo*[a]	*sapiens*[b,c]	789	481		1270
	Pan	*troglodytes*[d,e,b,f]	53	28	14	95
		paniscus[e,g]	9	9		18
	Gorilla	*gorilla*[d,e,b,f]	37	27		66
		beringei[d]	3	4	2	7
	Pongo	*pygmaeus*[d,e,b,f]	14	15	5	34
		abelli[d,f]	3	2		5
	Hylobates	*lar*[f]	39	34	7	80
	Symphalangus	*syndactylus*[b]	3	3	1	7
						Total = 1582
Cercopithecoidea	*Cercopithecus*	*albogularis*[h]	5	4		9
		mitis[b]		1		1
	Colobus	*guereza*[h,b]	8	9	3	20
	Macaca	*fascicularis*[f]	10	10		20
		nemestrina[b,f]	2	6		8
	Nasalis	*larvatus*[e]	3	5	2	10
	Papio	*cynocephalus*[h]	19	18		37
		hamadryas[h]	0	2		2
	Presbytis	*rubicunda*[f]	5	5		10
	Piliocolobus	*badius*[i]	2	11		13
	Theropithecus	*gelada*[j]		13		13
	Trachypithecus	*cristatus*[f]	5	5		10
						Total = 153
Ceboidea[b]	*Alouatta*	*caraya*	2	3	1	6
		palliata	3	2		5
	Aotus	*trivirgatus*	3	3		6
	Ateles	*fusciceps*	3	3		6
		geoffroyi	1	1	2	4
	Callicebus	*moloch*	3	2		5
	Callimico	*goeldii*	2	3		5
	Callithrix	*jacchus*	2	2		4
	Cebuella	*pygmaea*	4	1		5
	Cebus[k]	*albifrons*	3	3		6
		apella	2	3		5
	Lagothrix	*lagotricha*	1	1	2	4
	Leontopithecus	*chrysomelas*	3	2		5
		rosalia	2	2	1	5
	Saguinus	*fuscicolis*	2	1		3
		labiatus	3	3		6
		oedipus	1	3		4
	Saimiri	*sciureus*	3	3		6
						Total = 90
						n total = 1825

[a] The modern human sample derives specimens from the following locations: Alaska, the Andaman Islands, Argentina, Arizona, Australia, Austria, Belgium, California, the Canary Islands, Chile, the Democratic Republic of the Congo, Ecuador, Egypt, England, France, Germany, Hawaii, Illinois, Italy, Japan, Kentucky, Madagascar, New Jersey, New Mexico, Ohio, Peru, Philippines, Russia, Scotland, South Africa, South Dakota, Sudan, Utah, and Washington.
[b] Specimens of the Smithsonian National Museum of Natural History, Washington, D.C.
[c] Data from the Goldman Osteometric Dataset (https://web.utk.edu/~auerbach/GOLD.htm).
[d] Specimens of the American Museum of Natural History.
[e] Specimens of the Cleveland Museum of Natural History.
[f] Specimens of the Museum of Comparative Zoology, Harvard, and data from Auerbach and Ruff (2004, 2006).
[g] Specimens of the Royal Museum of Central Africa, Tervuren.
[h] Specimens of the National Museums of Kenya, Nairobi.
[i] Specimens of the Powell-Cotton Museum, Birchington-on-Sea.
[j] Specimens of the Zurich Zoo, Zürich.
[k] We maintain the use of genus *Cebus* to describe both robust (i.e., *Sapajus*) and gracile capuchins to maximize potential intra-generic variability.

2.2. Data collection

Non-human anthropoid data were collected using Mitutoyo digital calipers and osteometric boards, or three-dimensional (3D) polygonal models created from laser surface scans of long bones with Landmark Editor v. 3.0 (Wiley, 2006). Additional data were provided by J.M. Plavcan, C.B. Ruff, and M. Tocheri. An error study was conducted on a subset of the sample to assess measurement consistency across sources (Table 2). Inter-observer error was minimal (<5%), with the exceptions of the tibia maximum length of *Pongo,* which had 3D inter-point distances that differed by less than 7%, and several *Gorilla* specimens that differed by approximately 10% from those measured directly on skeletons by the first author. The source of error could not be determined, so only *Gorilla* specimens directly measured by the first author were included in the analyses, and *Gorilla* measurements from 3D scans were omitted.

Maximum lengths, including epiphyses, were collected for the humerus (HML), radius (RML), femur (FML), and tibia (TML). Measurement protocols follow those described in the documentation for the Goldman Osteometric Dataset (Auerbach and Ruff, 2004, 2006), where linear measurements extend from the most superior or proximal to the most inferior or distal points of each long bone (e.g., the distance from the most superior point of the humeral head to the most distal point of the medial projection of the humeral trochlea; Table 3). Only specimens with well-preserved representatives of the long bones of interest were included, and where possible measurements were collected from the right side of the skeleton.

2.3. Size adjustment

Historically, four dimensionless indices, intermembral (IMI; [HML + RML]/[FML + TML] × 100), humerofemoral (HFI; HML/FML × 100), brachial (BRI; RML/HML × 100), and crural (CRI; TML/FML × 100) have been used to investigate the relationship between primate postcranial proportions and positional behavior (e.g., Erikson, 1963; Fleagle, 1988; Hunt, 2016; Mollison, 1910; Napier, 1967; Napier and Napier, 1967). Napier and Napier (1967; see also Jungers, 1985; Granatosky, 2018) interpreted the greater sizes of the forelimbs relative to hindlimbs (i.e., larger intermembral or humerofemoral indices) and longer forearms relative to arms (i.e., larger brachial index) to indicate a greater reliance on forelimb-dominated locomotor patterns, while the opposite is true of larger hindlimbs (i.e., smaller intermembral or humerofemoral indices) and larger legs relative to thighs (i.e., larger crural index. The traditional limb indices do not include body size information, so to account for inter-individual body size differences and identify relative size differences of individual limb segments, we also size-adjusted individual linear measurements using the geometric mean (GM; Gaulin, 1979; Sailer et al., 1985; Cant, 1987; Doran, 1993a; Hunt et al., 1996). The GM integrates size data from the four linear variables into a single GM variable reflecting overall size (Jungers et al., 1995; Mosimann, 1970). Each linear measurement is divided by the GM, which is the nth root of the product of n input variables, to generate four dimensionless Mosimann-adjusted shape ratios: humerus maximum length/geometric mean (gHUM), radius maximum length/geometric mean (gRAD), femur maximum length/geometric mean (gFEM), and tibia maximum length/geometric mean (gTIB). In this study, we utilize both the traditional indices and the Mosimann shape ratios to search for evidence of distinctive patterns of covariance. Abbreviations and their definitions are provided in Table 3.

Table 2
Percent inter-observer error.

Comparates	*n*	% HML[a]	% RML[a]	% FML[a]	% TML[a]
Hylobates lar First author–Ruff[b]	7	0.43	0.60	0.30	0.38
Hylobates lar	7	0.60	0.42	0.65	0.91
Pan troglodytes	5	0.60	0.76	0.88	0.99
Pongo pygmaeus	3	0.65	3.76	4.82	6.80
First author–3D[c]					
Gorilla gorilla[d]	7	11.77	11.15	10.40	10.30
Hylobates lar	7	1.11	0.42	1.56	0.94
Pan troglodytes	5	2.41	2.56	3.13	1.17

Abbreviations: 3D = three-dimensional; HML = humerus maximum length; RML = radius maximum length; FML = femur maximum length; TML = tibia maximum length.

[a] Values are species averages, per variable, as percentages.

[b] Data from Ruff (2002).

[c] Data collected by the first author from 3D scans provided by Plavcan.

[d] Due to the error associated with *Gorilla,* only measurements collected by the first author by hand are used herein.

Table 3
Measurement definitions from published literature.

Linear measurements[a]	Definition
Humerus maximum length (HML)	The distance from the most superior point of the humeral head to the most distal point of the medial projection of the trochlea.
Radius maximum length (RML)	The distance between the most superior point on the radial head to the most distal point of the styloid process.
Femur maximum length (FML)	The distance from the most superior point on the femoral head to the most inferior aspect of the medial condyle.
Tibia maximum length (TML)	The distance from most superior point of the intercondylar eminence to the most distal aspect of the medial malleolus.
Traditional indices[b]	
Intermembral index (IMI)	(HML + RML)/(FML + TML) × 100
Humerofemoral index (HFI)	HML/FML × 100
Brachial index (BRI)	RML/HML × 100
Crural index (CRI)	TML/FML × 100
Mosimann-adjusted ratios[c]	
Geometric humerus (gHUM)	HML/GM
Geometric radius (gRAD)	RML/GM
Geometric femur (gFEM)	FML/GM
Geometric tibia (gTIB)	TML/GM

Abbreviations: HML = humerus maximum length; RML = radius maximum length; FML = femur maximum length; TML = tibia maximum length; IMI = intermembral index; HFI = humerofemoral index; BRI = brachial index; CRI = crural index; gHUM = geometric humerus; gRAD = geometric radius; gFEM = geometric femur; gTIB = geometric tibia; GM = geometric mean.

[a] Linear measurements follow protocols defined in the Goldman Osteometric Dataset (https://web.utk.edu/~auerbach/GOLD.htm; Auerbach and Ruff, 2004, 2006).

[b] Traditional indices are calculated following Schultz (1930).

[c] Mosimann-adjusted ratios are calculated according to Mosimann (1970).

2.4. Data analysis

We test the hypothesis that limb length proportions differentiate extant anthropoid taxa by using a series of one-way analyses of variance (ANOVA) and Tukey's post-hoc honestly significant difference (HSD), using an alpha value of $p < 0.05$ at all taxonomic levels. Boxplots (function 'qplot' from the 'ggplot2' package v. 3.4.4; Wickham, 2009) illustrate the ranges of variation for each of the four traditional limb indices at the genus level. Tukey's HSD is performed with function 'TukeyHSD' from the 'multcomp' package v. 1.4–14 (Hothorn et al., 2008). These and all subsequent analyses are performed in R v. 3.4.4 (R Core Team, 2017).

We used principal components analysis (PCA) to explore intra- and inter-taxon phenotypic patterns in a multivariate context. Because it summarizes multidimensional data visually, PCA traditionally has been used to study complex shape variation among single bones (e.g., humerus, mandible; Yang et al., 2006; Lague et al., 2008) or compound skeletal elements (e.g., cranium, pelvis; Franklin et al., 2006; Bonneau

et al., 2012; Hammond et al., 2018), but it has been used less frequently in primates to explore variation within and among multiple postcranial elements (but see Holliday, 1997; Young, 2003; Kivell et al., 2013). Principal component analyses were performed using the 'prcomp' function in base R and resulting biplots were generated with 'ggplot2'.

To quantify the correlations between limb proportions and locomotor behaviors, we performed three two-block partial least squares (2B-PLS) analyses on subsets of the sampled taxa using the 'two.b.pls' function of the package 'Geomorph' v. 4.0.6 (Adams et al., 2017). Two-block partial least squares constructs linear combinations of pairs of variables from one dataset (e.g., anthropoid limb proportions) such that a second dataset (e.g., percent locomotor mode) accounts for as much of the covariance between the original pairs of variables as possible (Rohlf and Corti, 2000). Two-block partial least squares does not assume *a priori* directional relationships among variables, so it is particularly appropriate for investigating and describing covariance between behavior and limb proportions without assuming one is necessarily predicated on the other (Adams and Felice, 2014).

The first two 2B-PLS analyses use arboreal locomotion data for adult specimens of the 24 species belonging to 18 genera, including *Homo sapiens*, listed in Table 4. Locomotor data for the Ceboidea were combined from several sources listed in Table 4. Cercopithecoid data were collected mainly from Hunt (2016; also see Gebo and Chapman, 1995; Cant et al., 2001 for original data). Hominoid locomotor data were collected primarily from Almécija et al. (2015; also see Thorpe and Crompton, 2006, and references therein for original data). Congener data were substituted for monkeys where locomotor data were unpublished, and averages were used when a genus had multiple reports: *Alouatta caraya* (*A. palliata*); *Presbytis rubicunda* (average *P. obscura* and *P. melalophos,* per Hunt, 2016); *Cebus albifrons* (*Cebus capucinus); Cercopithecus albogularis* (average of *Cercopithecus* species per Hunt, 2016); *Colobus guereza* (average of *C. guereza* per Hunt, 2016); *Papio cynocephalus* (average of semi-terrestrial *Papio* species per Hunt, 2016); *Piliocolobus badius* (average of *P. badius* per Hunt, 2016). *Macaca nemestrina* (*M. fuscata*), *Saimiri sciureus* (*Saimiri boliviensis*). We recognize that this approach may influence our results where monkeys are concerned, but our current focus is on hominoids, so we have included these data to increase our comparative sample. We will update and expand upon these results in a forthcoming monkey-focused study.

The six arboreal locomotor categories used by Almécija et al. (2015) are used as locomotor block variables: quadrupedal/tripedal walking (QTW), vertical climbing and descent (VCD), bipedal walking (BW), orthograde clamber and transfer (OCT), brachiation and forelimb swing (BFS), and drop and leap (DL). These arboreal locomotor categories are similar to those used in the other data sources detailed in Table 4, which allow for the inclusion of terrestrial *H. sapiens* as 100% bipedal walkers (Collard and Wood, 1999; Prost, 1980; Rose, 1984). We consider those behaviors that require a relatively greater weight bearing or propulsive use of the forelimbs than the hindlimbs to constitute forelimb-dominated activities (i.e., VCD, OCT, BFS); the inverse constitute hindlimb dominated activities (i.e., QTW, BW, DL). Some locomotor behaviors like bipedal walking are quantitatively different in monkeys, apes, and humans, which warrants care when interpreting results. For consistency and to include modern humans in analyses, we have maintained the approaches of Almécija et al. (2015) and Hunt (2016) in disregarding such mechanical differences.

For the first two 2B-PLSs, postural categories were excluded from all collected data, where applicable. Each behavioral category used was converted to a percentage of the sum of all behavioral data for each taxon. No other transformations were performed. We have attempted to accurately match behavioral categories described by multiple, independent authors, but some incongruity is expected, particularly for behaviors that are unique among different primate groups. For example, to increase comparability between publications, Hunt's (2016) 'clamber' and 'transfer' categories were combined into a single orthograde clamber and transfer (OCT) category, and Hunt's 'brachiate', and 'suspensory' categories were combined in the brachiation and forelimb swing (BFS) category, which includes use of the tail in the suspensory atelids. All other categorical adjustments are described in Table 4.

A third 2B-PLS is performed using the 18 positional modes (Table 5) published in Hunt (2016) and a subset of 11 catarrhines species: *Cercopithecus albogularis, Cercopithecus mitis, Colobus guereza, Hylobates lar, Pan troglodytes, Papio cynocephalus, Pongo abelli, Pongo pygmaeus, Presbytis rubicunda, Piliocolobus badius,* and *Symphalangus syndactylus*. Modern humans are not included in this 2B-PLS because the frequencies with which they use postural behaviors like 'sit', 'lie', and 'squat' are unavailable (see Hunt, 2016). Forelimb-dominated behaviors include 'cling', 'armhang', 'ips.susp', 'vert.climb', 'brach', 'clamber', 'susp', 'transf', and 'sway'; hindlimb-dominated behaviors include 'sit', 'squat', 'b.stand', 'q.walk', 'leap', 'q.run', 'b.walk'. The 'lie' posture is

Table 4
Percent locomotor behavior based on compiled published literature.[a]

Taxa	QTW	VCD	BW	OCT	BFS	DL
Ceboidea						
Alouatta caraya[b]	85.7	10.9	0.0	0.0	0.0	3.4
Ateles geoffroyi[i]	51.5	9.6	2.3	0.0	22.9	13.7
Cebuella pygmaea[e]	20.6	62.4	0.0	4.3	0.0	12.7
Cebus albifrons[b]	81.1	13.5	0.0	0.0	0.0	5.3
Cebus apella[f]	73.4	18.2	0.0	0.0	0.0	8.4
Lagothrix lagothricha[d,k]	35.3	17.1	0.1	36.5	6.1	4.9
Saimiri sciureus[i]	69.5	5.1	0.6	0.0	0.4	24.5
Cercopithecoidea						
Cercopithecus albogularis[h,j]	61.8	26.6	0.0	0.0	0.0	11.6
Cercopithecus mitis[h,j]	53.4	34.7	0.0	0.0	0.5	11.5
Colobus guereza[h,j]	47.6	20.8	0.8	0.0	0.6	30.2
Macaca fascicularis[c]	82.8	9.0	0.2	1.4	0.0	6.6
Macaca nemestrina[g]	86.5	0.0	0.0	0.0	0.0	13.5
Papio cynocephalus[j]	98.4	1.2	0.0	0.0	0.0	0.3
Presbytis rubicunda[j]	53.4	13.3	0.0	0.0	0.1	33.3
Piliocolobus badius[h,j]	42.1	32.8	0.3	0.0	0.7	24.1
Hominoidea						
Homo sapiens	0.0	0.0	100.0	0.0	0.0	0.0
Pan troglodytes[k]	35.3	48.0	6.9	4.9	4.9	0.0
Pan paniscus[k]	25.7	42.6	0.8	20.5	7.2	3.2
Gorilla gorilla[k]	20.4	51.6	5.4	18.3	3.2	1.1
Gorilla beringei[k]	50.0	37.7	1.9	5.7	4.7	0.0
Pongo abelli[j]	19.8	27.4	7.6	21.7	23.6	0.0
Pongo pygmaeus[k]	15.2	24.0	5.9	37.1	17.2	0.6
Hylobates lar[k]	0.5	23.7	3.7	0.0	60.1	12.0
Symphalangus syndactylus[j,k]	24.0	5.9	0.6	44.1	1.4	24.0

Abbreviations: QTW = quadrupedal/tripedal walking; VCD = vertical climbing and descent; BW = bipedal walking; OCT = orthograde clamber and transfer; BFS = brachiation and forelimb swing; DL = drop and leap.

[a] Percent locomotor behavior is calculated as the frequency of use of each sampled locomotor behavior as a percentage of the sum of all sampled locomotor behaviors. Congener data were substituted if locomotor data were not available for a given species: *Alouatta caraya* (*Alouatta palliata*); *Presbytis rubicunda* (average *Presbytis obscura* and *Presbytis melalophos,* per Hunt, 2016); *Cebus albifrons* (*Cebus capucinus); Cercopithecus albogularis* (average of *Cercopithecus* species per Hunt, 2016); *Colobus guereza* (average of *Colobus guereza* per Hunt, 2016); *Papio cynocephalus* (average of semi-terrestrial *Papio* species per Hunt, 2016); *Piliocolobus badius* (average of *Piliocolobus badius* per Hunt, 2016). *Macaca nemestrina* (*Macaca fuscata*), *Saimiri sciureus* (*Saimiri boliviensis*).

[b] From Bezanson (2009): VCD comprises 'bimanual pull-up', 'flexed vertical climb', 'vertical scramble', 'pulse climb', 'cling'; BFS comprises 'tail/forelimb suspension'.

[c] From Cant (1988): averages travel and forage/feeding; OCT comprises 'vertical clambering'.

[d] From Cant et al. (2001): only arboreal, travel and feeding.

[e] From Youlatos (1999): combines forage, feeding, and travel; QTW is only quadrupedal walk; OCT is 'vertical grasp' and 'claw climb'.

[f] From Wright (2007).

[g] From Chatani (2003).

[h] From Gebo and Chapman (1995).

[i] From Fontaine (1990).

[j] From Hunt (2016).

[k] From Almécija et al. (2015). See also Thorpe and Crompton (2006).

Table 5
Percent positional behavior from Hunt (2016).[a,b]

Taxa	sit	lie	q.stand	squat	cling	b.stand	armhang	ips.susp	q.walk	vert.climb	leap	q.run	b.walk	brach	clamber	susp	transf	sway
Cercopithecoidea																		
Cercopithecus albogularis	32.9	0.6	12.2	0.0	8.4	0.0	0.0	0.0	26.6	10.7	5.7	6.0	0.0	0.0	0.0	0.1	0.0	0.0
Cercopithecus mitis	30.1	0.9	15.1	0.0	0.0	1.4	0.0	0.0	28.6	18.5	5.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Colobus guereza	31.1	2.9	1.4	0.0	0.0	0.0	0.0	0.0	26.3	10.3	27.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Papio cynocephalus	57.1	3.2	14.5	0.2	0.3	0.0	0.0	0.0	23.2	0.5	0.2	0.4	0.0	0.0	0.0	0.0	0.0	0.0
Piliocolobus badius	30.7	0.3	2.7	0.0	0.0	0.3	0.0	0.0	24.4	23.0	17.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Presbytis rubicunda	22.1	0.0	0.0	0.0	0.0	0.0	2.1	0.0	36.8	10.4	26.5	0.0	0.0	2.1	0.0	0.0	0.0	0.0
Hominoidea																		
Pan troglodytes	62.4	12.1	2.5	0.7	0.3	0.3	4.4	0.0	16.1	0.9	0.0	0.3	0.1	0.1	0.0	0.1	0.2	0.0
Pongo abelii	41.8	8.6	2.9	0.3	0.1	4.3	8.0	5.8	5.2	7.2	0.0	0.0	2.0	1.7	4.0	4.5	1.7	1.7
Pongo pygmaeus	55.6	4.0	1.1	0.0	0.8	3.8	12.4	6.9	1.3	2.4	0.0	0.0	0.5	1.8	6.5	0.8	0.0	3.1
Hylobates lar	29.1	0.0	0.0	0.0	0.0	0.0	46.7	0.0	0.0	8.6	2.3	0.0	1.3	12.4	0.0	0.0	0.0	0.0
Symphalangus syndactylus	20.4	0.0	0.0	0.0	0.0	0.0	32.9	0.0	0.0	25.4	1.5	0.0	2.2	17.7	0.0	0.0	0.0	0.0

Abbreviations: q.stand = quadrupedal standing; b.stand = bipedal standing; arm.hang = arm hanging; ips.susp = ipsilateral suspension; q.walk = quadrupedal walking; vert.climb = vertical climbing; q.run = quadrupedal running; b.walk = bipedal walking; brach = brachiation; susp = bimanual suspension; transf = transfer.

[a] See Hunt (2016) for descriptions of each category.

[b] Percent positional behavior is calculated as the frequency of use of each sampled positional behavior as a percentage of the sum of all sampled positional behaviors: Congener data were substituted if locomotor data were not available for given species: *Presbytis rubicunda* (average *Presbytis obscura* and *Presbytis melalophos*); *Cercopithecus albogularis* (average of *Cercopithecus* species); *Colobus guereza* (average of *C. guereza*); *Papio cynocephalus* (average of semi-terrestrial *Papio* species); *Piliocolobus badius* (average of *P. badius*).

considered neutral in terms of limb-dominance (Hunt et al., 1996). The results from Hunt's positional data are compared against the results of the compiled arboreal locomotion data to better understand how selection and standardization of behavioral categories affect the outcomes of the analyses.

Phylogenetic comparative methods (PCMs) were used to test the phylogenetic signal in Mosimann-adjusted shape variables and traditional limb indices. The datasets analyzed included multiple taxa from three anthropoid superfamilies. Given that species are not evolutionarily independent, accounting for any effects due to phylogeny is crucial to avoid confounding effects of shared ancestry (Blomberg and Garland, 2002; Collyer et al., 2000, 2015). We obtained the current consensus of the phylogenetic relationships among the complete anthropoid dataset (SOM Fig. S1) from a consensus tree based on molecular data downloaded from 'The 10KTrees Project' v. 3 (Arnold et al., 2010; https://10ktrees.nunn-lab.org/). The nearest available sister species are used as substitutes for species not included in 10K Trees (e.g., *Saguinus midas* represents *Saguinus labiatus*; *Presbytis melalophos* represents *Presbytis rubicunda*). The obsolete generic designation of *Cebuella* (*Callithrix*) *pygmaea* is maintained in these analyses to distinguish it from *Callithrix jacchus,* due to their large body size differences. The phylogram in SOM (Figure S1) is generated using Mesquite v. 3.61 (Maddison and Maddison, 2019).

Phylogenetic signal is estimated for both traditional indices and Mosimann-adjusted shape ratios using Blomberg's K (Blomberg et al., 2003), a method previously used to study phylogenetic effects on taxic diversity and ecological communities (Flynn et al., 2011; Kraft et al., 2007; Swenson et al., 2012), and the degree and pattern of phylogenetic signals in the long bone structure of quadrupedal primates (O'Neill and Dobson, 2008). Blomberg's K, which assumes a Brownian motion model of trait evolution, estimates phylogenetic signal as the ratio of the mean squared error of the tree tip data (MSE_0) measured from the phylogenetic corrected mean for a given continuous variable, and the mean squared error of the variance-covariance matrix derived from the given phylogeny (Blomberg et al., 2003; Münkemüller et al., 2012). A Blomberg's K value of 1.0 indicates that the variance of a continuous trait among taxa exactly matches that expected under a Brownian model and the given phylogenetic tree, whereas $K < 1.0$ indicates less resemblance than expected, and $K > 1.0$ indicates more resemblance than expected, among closely related taxa. A multivariate version of Blomberg's K (hereafter called multiple K; Adams, 2014) allows for the estimation of phylogenetic signal using multiple continuous traits simultaneously. We use the univariate Blomberg's K to understand how individual limb proportions contribute to overall phylogenetic signal, and multiple K to understand how they interact. Blomberg's K and multiple K, and their associated *p*-values (derived at 10,000 permutations) are calculated using the 'physignal' function of the 'Geomorph' package.

3. Results

3.1. Differentiating extant anthropoids by means of their limb proportions

Our results support the hypothesis that traditional limb proportions separate extant anthropoid taxa. Boxplots illustrating genus-level variation for each of the four traditional indices are provided in Figure 1. Visual inspection reveals differences in limb proportions between modern humans and other anthropoids consistent with those reported in previous studies (Schultz, 1930; Shea, 1981; Jungers, 1985; Richmond et al., 2002). Asian apes have the highest IMI values (i.e., *Pongo, Hylobates,* and *Symphalangus*: 125–140%), followed by the African apes (i.e., *Gorilla* and *Pan*: 100–120%). Asian and African apes are separated from *H. sapiens* by monkeys. *Pan*, which has nearly identical values for IMI and HFI (ca. 100–110%), has the lowest indices among non-human hominoids. The BRI values of atelids overlap with *Pongo, Hylobates,* and most cercopithecoids. *Presbytis* has the highest BRI values among cercopithecoids. Crural ratio broadly separates anthropoid

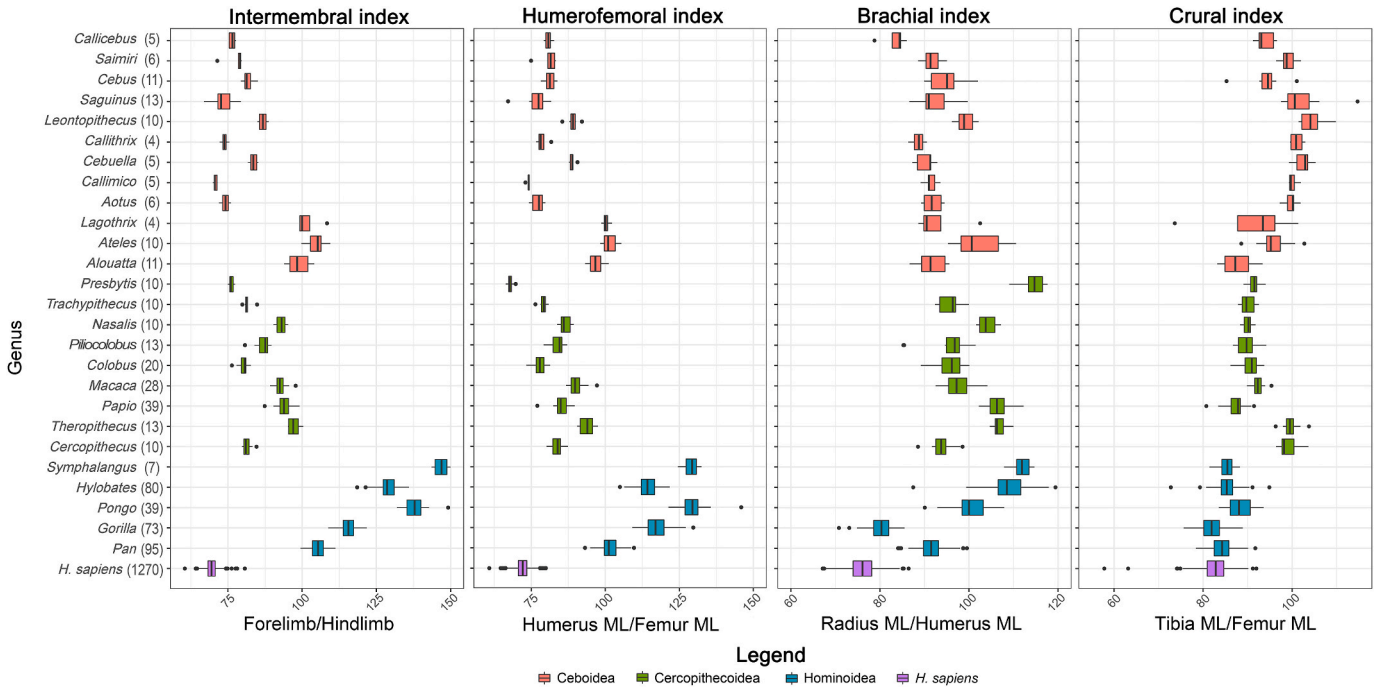


Fig. 1. Box plots of anthropoid traditional indices at the genus level. For most indices, *H. sapiens* are separated from the Hominoidea, with monkeys intermediate. Sample sizes are in parentheses beside genus names.

superfamilies. Monkeys have higher CRI values and ranges (ca. 85–05% for Cercopithecoidea, and ca. 90–110% for Ceboidea) than extant hominoids (ca. 75–85% for modern humans and African apes, and ca. 80–95% for Asian apes). For IMI and HFI, modern humans are more dissimilar to their nearest relatives than they are to any other genera, but for BRI they overlap with *Gorilla*, and for CRI they are nearest to the other African apes. Small sample sizes for some monkeys may underrepresent their true variation, but it is worth noting that the ranges of variation for *H. sapiens* variables do not greatly exceed those of other taxa, despite the large sample size and demographic diversity of the human data. However, in box plots of the crural index, two individuals from the Goldmann Osteometric Dataset, of respective Japanese and North American origin, stand out for their exceedingly low crural indices.

To quantify the differences evident in boxplots, ANOVA with post-hoc Tukey's HSD multiple pairwise comparisons were performed on the complete anthropoid sample, and on hominoids, cercopithecoids, and ceboids at select taxonomic levels: superfamily, genus, and species levels for Anthropoidea; genus and species levels for Hominoidea, Cercopithecoidea, and Ceboidea. Below the superfamily level, hominoids are most dissimilar, regardless of the class of variables used. Non-significant differences among hominoids usually involved pairs of the Asian apes (*Pongo, Hylobates,* and *Symphalangus*). Results of these analyses are summarized below and listed in Table 6.

In general, pairwise differences indicate IMI and HFI as the variables that best differentiate the extant groups, regardless of their taxonomic

Table 6
Tukey's honestly significant difference (HSD) results per selected taxonomic levels. Percentages of unique pairwise HSD tests that result in $p < 0.05$ within taxonomic groupings at superfamily, genus, and species levels.[a]

Variable	Anthropoidea (superfamily)	Anthropoidea (genus)	Anthropoidea (species)	Hominoidea (genus)	Hominoidea (species)	Cercopithecoidea (genus)	Cercopithecoidea (species)	Ceboidea (genus)	Ceboidea (species)
IMI	100.0%	86.6%	81.7%	100.0%	82.6%	86.1%	69.6%	80.0%	73.4%
HFI	100.0%	82.1%	75.2%	90.0%	76.1%	83.3%	60.8%	77.8%	66.9%
BRI	83.3%	67.3%	53.6%	90.0%	76.1%	66.7%	54.4%	44.4%	27.9%
CRI	100.0%	66.2%	60.3%	80.0%	37.0%	66.7%	49.4%	44.4%	23.4%
All traditional indices	83.3%	79.5%	72.8%	100.0%	82.2%	75.0%	65.2%	73.3%	63.4%
gHUM	83.3%	68.4%	64.5%	90.0%	68.9%	69.4%	47.0%	66.7%	50.3%
gRAD	100.0%	84.9%	70.1%	90.0%	64.4%	83.3%	72.7%	80.0%	69.3%
gFEM	83.3%	74.6%	63.4%	90.0%	75.5%	86.1%	69.7%	68.9%	57.5%
gTIB	100.0%	82.6%	77.9%	100.0%	77.8%	91.7%	77.3%	66.7%	59.5%
All Mosimann-adjusted ratios	83.3%	61.8%	47.6%	100.0%	80.0%	86.1%	62.1%	64.4%	57.5%
PC1	100.0%	85.8%	80.8%	100.0%	82.6%	86.1%	67.1%	82.2%	73.4%
PC2	100.0%	61.4%	56.6%	90.0%	67.4%	72.2%	55.7%	53.3%	22.1%
Number of comparisons	6	351	820	10	46	36	79	45	154
Total compasisons	1468								

Abbreviations: IMI = intermembral index; HFI = humerofemoral index; BRI = brachial index; CRI = crural index; gHUM = geometric humerus; gRAD = geometric radius; gFEM = geometric femur; gTIB = geometric tibia; GM = geometric mean; PC = principal component.

[a] Comparisons above used Mosimann-adjusted ratios, traditional indices, PC1, and PC2 of traditional indices. Number of comparisons is the number of pairs of taxa analyzed by Tukey's HSD. Post-hoc comparisons revealed small differences in limb proportions within the ceboids and within the cercopithecoids at their species levels, challenging previous assumptions about the homogeneity of these groups.

level. Interesting caveats include the anthropoid superfamilies, all (100%) of which can be differentiated when using CRI as well as they can be when using IMI or HFI; and hominoid genera and species, which are differentiated using BRI as well as HFI (90% of genera; 76% of species). Intermembral index (100% of genera; and 83% of species) outperforms both BRI and HFI. Cercopithecoid genera are best separated by gTIB (91.7%), and second best by IMI, gFEM, combined Mosimann-adjusted ratios, and PC1 (86.1%). Ceboid genera are best separated using PC1 (82.2%), and second best by IMI and gRAD (80%). Using Mosimann ratios, distal elements (gRAD and gTIB) best differentiate groups at nearly all taxonomic levels (Table 6). Notable exceptions include hominoid species, which are best differentiated using size-adjusted lower limb elements (gFEM and gTIB). Principal component 1 always performs at a percentage between IMI and HFI, but PC2 underperforms with respect to most other indices. In post-hoc tests of combined traditional indices and combined Mosimann ratios, we found similar patterns differentiating 100% of hominoid genera, and approximately 80% of anthropoid superfamilies and hominoid species.

The multivariate spaces defined by the first two PC axes of traditional indices and of Mosimann-adjusted shape ratios are very similar, where the first two principal components explain approximately 98% of total variance (Fig. 2). In both PC plots, hominoid genera form three partially-overlapping groups comprising African apes, Asian apes, and modern humans, whereas the monkey superfamilies show considerable overlap with each other, and most are located between modern humans and African apes on PC1. Loadings indicate the direction and scale of each variable on the total variance of each principal component (Table 7).

In PCAs of traditional indices, PC1 explains 89% of the observed variance, and PC2 explains 9%. Between-limb relationships are best captured by PC1. Intemembral index loads most heavily on PC1, and CRI loads least. Within-limb relationships are most captured by PC2. On PC2, BRI loads most heavily in the negative direction, and HFI in the positive direction.

In PCAs of Mosimann plots, forelimb (gHUM and gRAD, toward the negative end) versus hindlimb (gFEM and gTIB, toward the positive end) lengths are captured on PC1, and distal (gRAD and gTIB, toward negative) versus proximal (gHUM and gFEM, toward positive) element lengths on PC2. On PC1, gFEM explains the greatest variance in the positive direction, and gRAD explains the greatest variance in the negative direction. Principal component 1 explains 91% of the variance, and PC2 explains 7% of the variance. In both traditional index and Mosimann PCA analyses, PC3 and PC4 explain a combined 2% of total variance and are not reported on further here.

Table 7
Principal component (PC) analysis loadings and eigenvalues. The direction and scale of each variable (loadings), and proportions of variance (eigenvalues) explained by each principal component.

Traditional indices[a]	PC1	PC2	PC3	PC4
IMI	−0.736	0.136	0.216	0.628
HFI	−0.590	0.333	−0.363	−0.639
BRI	−0.328	−0.839	0.306	−0.308
CRI	−0.050	−0.408	−0.853	0.321
Eigen values[b]	89.3	8.6	2.1	0.0
Mosimann ratios[a]	**PC1**	**PC2**	**PC3**	**PC4**
gHUM	−0.257	0.732	−0.393	0.494
gRAD	−0.634	−0.353	0.432	0.535
gFEM	0.594	0.298	0.644	0.380
gTIB	0.423	−0.502	−0.494	0.570
Eigen values[b]	91.2	6.8	1.9	0.2

Abbreviations: IMI = intermembral index; HFI = humerofemoral index; BRI = brachial index; CRI = crural index; gHUM = geometric humerus; gRAD = geometric radius; gFEM = geometric femur; gTIB = geometric tibia; GM = geometric mean.

[a] Loadings indicate the extent to which each original variable contributes to each PC.

[b] Together, PC1 and PC2 explain more than 95% of total variance in each analysis.

3.2. Linking behavior and limb proportions

Results of 2B-PLS analyses confirm that significant correlations exist between behavioral repertoires, and both traditional indices and Mosimann-adjusted shape ratios, but their relationships are complex and do not always accord to predictions (Fig. 3). All 2B-PLS scores are provided in Table 8. The 2B-PLS results indicate that an increase in climbing, suspensory, and orthograde behaviors are most associated with relatively longer forelimbs (e.g., some atelids, and all hominoids), or relatively longer distal forelimb elements (e.g., *Ateles*, *Presbytis*, and the Asian hominoids), while leaping/dropping, and bipedalism are most associated with relatively longer hindlimbs (e.g., *Homo, Cercopithecus, Macaca, Presbytis*, and most non-atelid New World monkeys), or

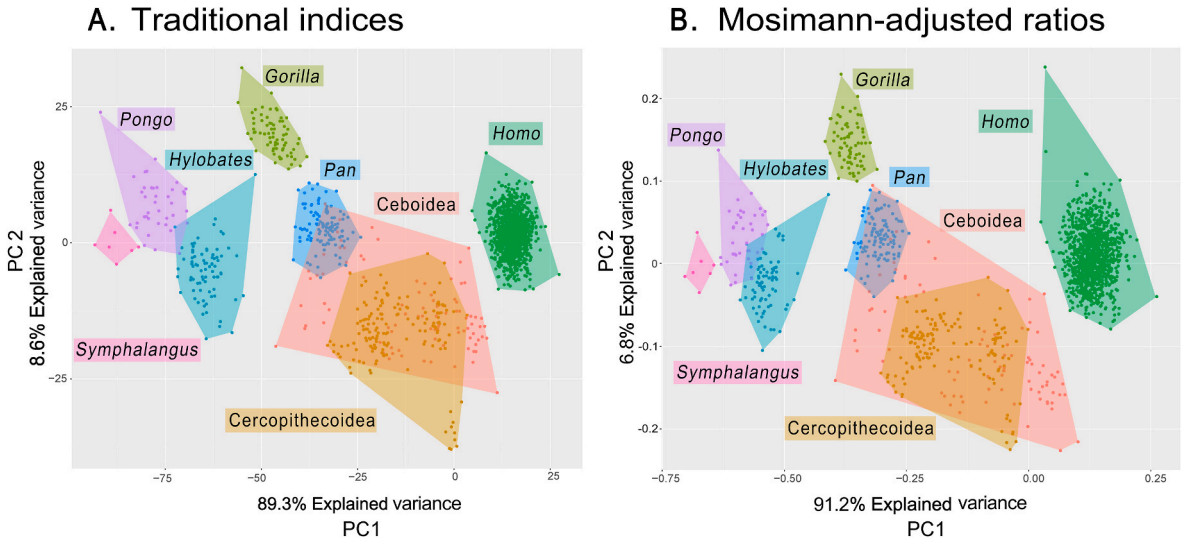


Fig. 2. Principal components analysis (PCA) of (A) traditional indices, and (B) Mosimann-adjusted shape ratios. The first two PCs explain approximately 98% of total variance. A small region of overlap between *Pongo* and *Hylobates* notwithstanding, hominoid genera are separated in PCA space with virtually no inter-generic overlap. However, *Pan* overlaps markedly with members of the Atelidae. On PC1, monkeys fill the gap between *Homo sapiens* and African apes, overlapping with all. Loadings and eigenvalues are provided in Table 7. A version of plot (B) with hulls indicating monkey genera is provided in SOM Fig. S2.

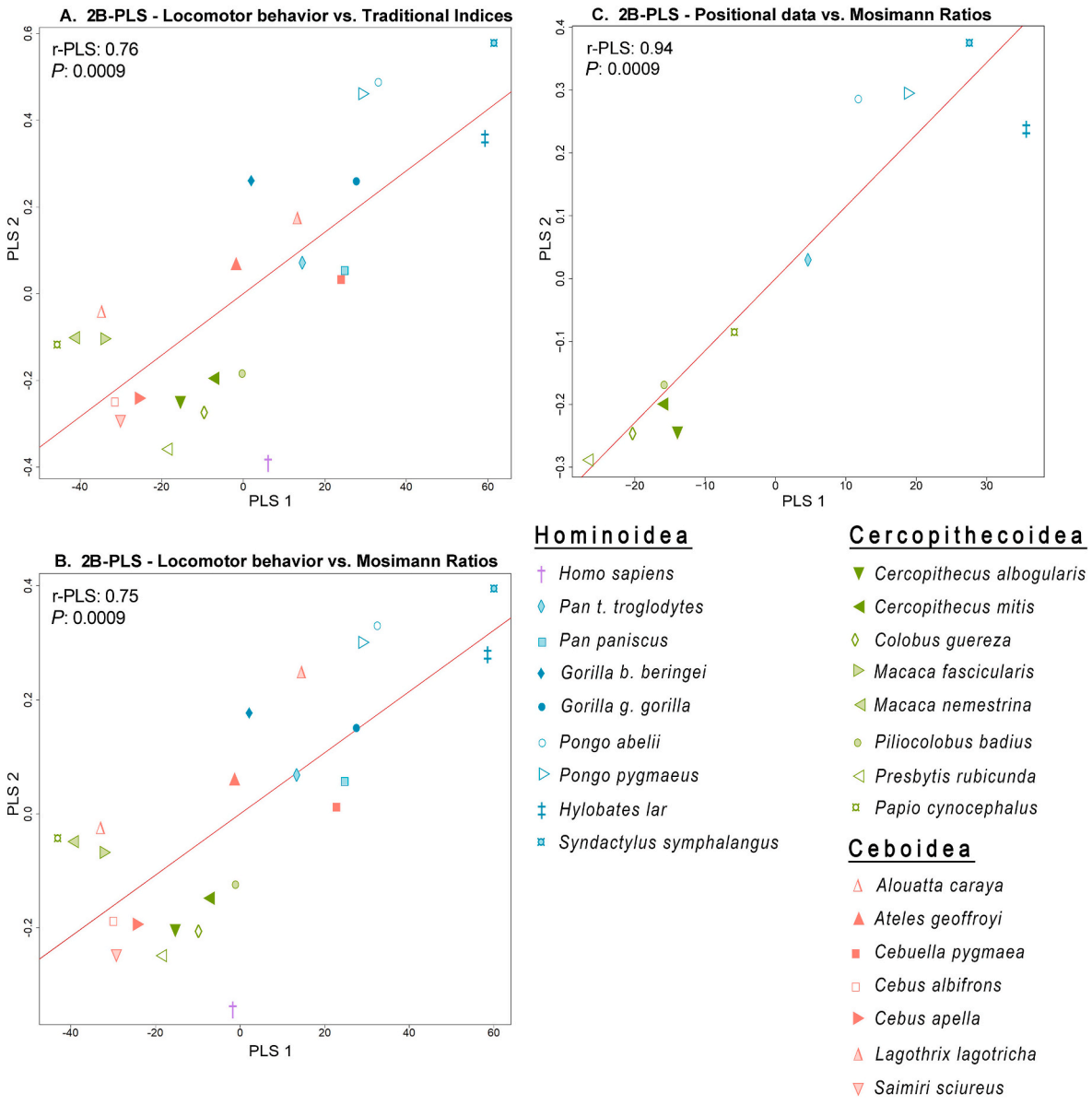


Fig. 3. Two-block partial least squares (2B-PLS) showing the positive and significant association between arboreal locomotor behavior and (A) traditional indices and (B) Mosimann-adjusted ratios. Positional data 2B-PLS using Mosimann-adjusted ratios and positional data from Hunt (2016; C).

relatively longer distal hindlimb elements (e.g., many New World monkeys). Some taxa would not fit these patterns. *Theropithecus*, despite its relatively elongated distal hind-limbs, does not practice leaping, dropping, or bipedalism, but is instead a committed terrestrial (hindlimb-dominated) quadruped whose congeneric predecessors may have behaved similarly (Schmidt, 2011).

Traditional indices Anthropoids exhibit significant correlations between traditional indices and locomotor mode (r-PLS = 0.76, $p < 0.001$). On the positive end of PLS 1, larger IMI, HFI, and BRI shape block variables are related to larger BFS, OCT, and VCD locomotor block variables. On the negative end, smaller IMI, HFI, larger CRI, and intermediate BRI are associated with QTW, DL, and BW. Negative PLS 2 locomotion block values comprise BW, VCD, and OCT which covary with larger HFI in the shape block. Positive PLS 2 locomotion block values are related to BFS, DL, and QTW, and correlated to greater BRI, CRI, and to a lesser extent, IMI in the shape block.

Mosimann-adjusted shape ratios Anthropoids exhibit significant correlations between Mosimann-adjusted shape ratios and frequency of arboreal locomotor behaviors (r-PLS = 0.75, $p < 0.001$). Positive PLS 1 values capture relatively longer forelimb elements (especially gRAD) and relatively shorter hindlimb elements (especially gTIB), which covary with BFS, OCT, and VCD. Negative PLS 2 variables are associated with relatively longer hindlimb elements (especially longer gTIB) and relatively shorter forelimb elements (especially shorter gRAD), and are related to QTW, DL, and BW. On PLS 2, negative shape values relate to longer distal segments relative to proximal elements, especially gRAD compared to gHUM, which covary with DL, BFS, and QTW. Positive shape variables include longer proximal and shorter distal elements, and covary with locomotion block values related to BW, VCD, and OCT (Table 8).

Positional data two-block partial least squares The 2B-PLS performed solely on the Hunt (2016) positional data and Mosimann ratios demonstrated a stronger correlation (r-PLS = 0.94, $p < 0.001$) than the preceding 2B-PLS analyses. Positive PLS 1values on the shape block correlate with longer forelimb versus hindlimb elements, especially gRAD versus gTIB, and are related to 'sitting' and forelimb-dominated

Table 8
Two-block partial least squares (2B-PLS) scores.

Locomotor behavior vs. traditional indices[a]					Positional data vs. Mosimann ratios[b]				
Shape block	PLS 1	PLS 2	PLS 3	PLS 4	Shape block	PLS 1	PLS 2	PLS 3	PLS 4
IMI	0.77	0.11	−0.29	0.56	gHUM	0.45	−0.65	−0.16	−0.60
HFI	0.62	−0.08	0.56	−0.55	gRAD	0.57	0.73	0.03	−0.37
BRI	0.04	0.89	−0.28	−0.37	gFEM	−0.43	0.03	0.72	−0.55
CRI	−0.15	0.45	0.72	0.50	gTIB	−0.53	0.22	−0.68	−0.46
Locomotor block					Positional block				
QTW	−0.69	0.30	0.34	0.31	sit	0.14	−0.71	0.39	0.27
VCD	0.29	−0.21	0.73	−0.26	lie	0.05	−0.24	0.12	−0.26
BW	−0.16	−0.67	−0.35	0.37	q.stand	−0.13	−0.07	−0.47	0.50
OCT	0.24	−0.23	−0.14	−0.24	squat	0.00	−0.01	0.01	0.00
BFS	0.55	0.52	−0.18	0.46	cling	−0.04	−0.03	−0.20	−0.05
DL	−0.24	0.30	−0.41	−0.65	b.stand	0.04	−0.06	−0.11	−0.13
Locomotor behavior vs. Mosimann-adjusted ratios					arm.hang	0.58	0.47	0.24	0.40
Shape block	PLS 1	PLS 2	PLS 3	PLS 4	ips.susp	0.07	−0.07	−0.12	−0.20
gHUM	0.50	0.63	−0.23	−0.55	q.walk	−0.66	0.14	0.11	0.43
gRAD	0.57	−0.45	0.63	−0.27	vert.climb	−0.03	0.18	−0.37	−0.28
gFEM	−0.35	0.58	0.74	0.03	leap	−0.38	0.29	0.54	−0.30
gTIB	−0.55	−0.26	−0.03	−0.80	q.run	−0.03	−0.02	−13.00	−0.24
Locomotor block					b.walk	0.04	0.01	−0.01	−0.05
QTW	−0.65	−0.35	−0.23	−0.35	brach	0.20	0.26	0.09	0.05
VCD	0.28	0.11	−0.81	0.17	clamber	0.06	−0.06	−0.10	−0.16
BW	−0.23	0.64	0.21	0.46	susp	0.03	−0.04	−0.04	−0.11
OCT	0.28	0.37	0.24	−0.73	transf	0.01	−0.02	−0.01	−0.04
BFS	0.56	−0.50	0.25	0.17					
DL	−0.24	−0.27	0.34	0.27					

Abbreviations: QTW = quadrupedal/tripedal walking; VCD = vertical climbing and descent; BW = bipedal walking; OCT = orthograde clamber and transfer; BFS = brachiation and forelimb swing; DL = drop and leap; IMI = intermembral index; HFI = humerofemoral index; BRI = brachial index; CRI = crural index; gHUM = geometric humerus; gRAD = geometric radius; gFEM = geometric femur; gTIB = geometric tibia; q.stand = quadrupedal standing; b.stand = bipedal standing; arm.hang = arm hanging; ips.susp = ipsilateral suspension; q.walk = quadrupedal walking; vert.climb = vertical climbing; q.run = quadrupedal running; b.walk = bipedal walking; brach = brachiation; susp = bimanual suspension; transf = transfer.

[a] Both traditional indices and Mossiman-adjusted ratios refer to 2B-PLS analyses based on compiled published literature.

[b] Positional data vs. Mosimann ratios refers to 2B-PLS analyses limited to only Hunt (2016) data.

behaviors 'arm hanging' and 'brachiating'. Negative values correlated with longer hindlimbs are related to 'quadrupedal walking', 'leaping', and 'quadrupedal standing'. Positive values on PLS 2 are mostly related to longer distal elements and shorter gHUM and longer gFEM, and correlate most with 'arm hanging', 'leaping', 'brachiating', 'vertical climbing', and 'quadrupedal walking'.

3.3. Phylogenetic signals in limb proportions and behavior

Blomberg's K and multiple K (Table 9) both supported the hypotheses of phylogenetic structure in limb proportions and, to a lesser extent, in behavioral repertoires. Blomberg's K detected significant phylogenetic signals for the combined, and most individual traditional indices (IMI, HFI, BRI, CRI), Mosimann-adjusted shape ratios (gHUM, gRAD,

Table 9
Phylogenetic signal in extant anthropoid limb proportions. Blomberg's K for individual Mosimann-adjusted shape ratios and traditional indices, as well as multiple K in bold for all Mosimann-adjusted shape ratios and all traditional indices, in several anthropoid groups.[a,b]

Variable	Anthropoids		Anthropoids (non-*Homo*)		Hominoids (non-*Homo*)		All Monkeys		Cercopithecoids		Ceboids	
	K	*p*	K	*p*	K	*p*	K	*p*	K	*p*	K	*p*
Traditional indices												
IMI	0.98	<0.01	1.65	<0.01	1.25	<0.01	1.06	<0.01	1.24	<0.01	2.03	<0.01
HFI	0.87	<0.01	1.40	<0.01	1.04	0.02	0.87	<0.01	0.97	<0.01	1.83	<0.01
BRI	0.61	0.51	0.64	<0.01	1.70	<0.01	0.80	<0.01	1.00	0.018	1.20	<0.01
CRI	1.11	<0.01	1.07	<0.01	1.46	<0.01	0.82	<0.01	0.72	0.149	1.60	<0.01
All Traditional indices	0.90	<0.01	1.35	<0.01	1.32	<0.01	0.92	<0.01	1.00	<0.01	1.81	<0.01
Mosimann-adjusted shape ratios												
gHUM	1.07	<0.01	0.92	<0.01	0.88	0.05	0.88	<0.01	0.95	0.01	1.90	<0.01
gRAD	0.78	<0.01	1.88	<0.01	1.70	<0.01	1.05	<0.01	1.31	<0.01	1.80	<0.01
gFEM	0.52	<0.01	1.49	<0.01	1.25	<0.01	0.72	<0.01	0.93	0.020	1.48	<0.01
gTIB	1.17	<0.01	1.26	<0.01	1.25	<0.01	1.20	<0.01	1.17	<0.01	2.13	<0.01
All Mosimann-adjusted shape ratios	0.85	<0.01	1.40	<0.01	1.21	<0.01	0.98	<0.01	1.09	<0.01	1.85	<0.01

Abbreviations: IMI = intermembral index; HFI = humerofemoral index; BRI = brachial index; CRI = crural index; gHUM = humerus ratio; gRAD = radius ratio; gFEM = femur ratio; gTIB = tibia ratio.

[a] *Homo* was excluded from two sets of tests to isolate the effect of the genus on results. K < 1 indicates less similarity, and K > 1 indicates greater similarity, among closely related taxa than expected under Brownian motion. Most groupings are at least as similar as expected under Brownian motion. K approaches or exceeds 1 in several tests.

[b] Definition of the groups: Anthropoids = total sample; Anthropoids (non-*Homo*) = all anthropoids excluding *Homo*; Hominoids (non-*Homo*) = all hominoids excluding *Homo*; Monkeys = all monkeys; Cercopithecoids = all Cercopithecoidea; Ceboids = all Ceboidea.

gFEM, gTIB), and major axes of variation (PC1, PC2). Exceptions include anthropoids that have a small Blomberg's K and non-significant phylogenetic signal for BRI ($K = 0.61$, $p = 0.51$), and cercopithecoids with a Blomberg's K approaching 1, but with a non-significant phylogenetic signal for CRI ($K = 0.72$, $p = 0.15$).

In contrast, fewer behavioral variables had non-random or significant phylogenetic signals (Table 10). In anthropoids, locomotor modes BFS, DL, QTW, and combined locomotor repertoires show significant phylogenetic signals approaching or exceeding expectations based on Brownian motion ($K \leq 0.80$, $p < 0.01$). In non-human hominoids, BFS has a striking phylogenetic signal ($K = 2.6$, $p < 0.01$). In cercopithecoids, DL, QTW and combined locomotor modes have strong phylogenetic signals ($K > 1$, $p < 0.05$). Ceboids demonstrate no phylogenetic signals in arboreal locomotor variables.

Using Hunt's (2016) positional data, non-*Homo* anthropoids have non-random Blomberg's K and significant *p*-values for several positional behaviors: q.stand ($K = 1.94$, $p = 0.02$), armhang ($K = 1.50$, $p < 0.01$), ips.suspend ($K = 1.30$, $p = 0.02$), q.walk ($K = 2.14$, $p < 0.01$), leap ($K = 1.66$, $p < 0.01$), brachiate ($K = 1.38$, $p < 0.01$), and overall combined positional behaviors ($K = 1.06$, $p < 0.01$). Only hominoids have non-random and significant phylogenetic signals for q.walk ($K = 1.09$, $p = 0.05$) and combined positional behaviors ($K = 1.16$, $p = 0.01$), and nearly significant values for armhang ($K = 1.48$, $p = 0.06$).

4. Discussion

4.1. How well do limb proportions differentiate anthropoid taxa?

Most hominoid, cercopithecoid, and ceboid taxa can be differentiated using some individual (particularly interlimb ratios) or combinations of limb proportions or major axes of variation, but the patterns are complex. Using interlimb indices and PCAs, combined monkey, cercopithecoid, and ceboid genera are differentiated at rates comparable or intermediate to hominoid species and genera. Below the superfamily level, hominoids are best differentiated using all variables and taxonomic levels tested.

The distribution of taxa in shape-space suggests that selection has diversified the limb length proportions of hominoids, some members of the Atelidae, and possibly *Presbytis*, from their nearest relatives. *Pongo* falls nearer to other arboreal Asian apes than to its nearest extant African ape relatives, while *Pan* and the Atelidae (especially *Alouatta* and *Lagothrix*) overlap markedly. Larson (1998) and Young (2003) both recognize homoplasy in *Ateles* and *Hylobates* postcranial measurements including forelimb articular dimensions, and the atelids' *Pan*-like inter-limb length proportions have been noted since at least Schultz (1930) and Napier and Napier (1967). The *Pan*-like atelid morphology evident in both PCAs separates them from other ceboids and may imply a striking selective effect of some locomotor patterns (e.g., suspensory behaviors: Larson, 1998; Hunt, 2016) on limb length proportions that are strongly mediated by phylogeny. This is supported by the

Table 10
Phylogenetic signal in behavioral variables. Blomberg's K for individual arboreal locomotion and positional behavior variables, as well as multiple K in bold for overall variables, in several anthropoid groups.[a,b]

Variable	Anthropoids		Anthropoids (non-*Homo*)		Hominoids (non-*Homo*)		Monkeys		Cercopithecoids		Ceboids	
	K	*p*	K	*p*	K	*p*	K	*p*	K	*p*	K	*p*
Arboreal locomotion variables												
QTW	0.80	<0.01	0.82	<0.01	0.60	0.16	0.66	0.03	1.43	0.02	1.01	0.20
VCD	0.41	0.06	0.63	<0.01	0.84	0.03	0.59	0.05	0.99	0.09	1.11	0.15
BW	0.27	0.64	0.37	0.16	0.31	0.57	0.41	0.52	1.20	0.08	0.65	0.57
OCT	0.34	0.24	0.36	0.18	0.51	0.17	0.39	0.48	0.55	0.69	0.63	0.57
BFS	1.09	<0.01	1.11	<0.01	2.59	<0.01	0.53	0.25	0.31	0.81	0.84	0.40
DL	0.92	<0.01	0.88	<0.01	0.69	0.08	0.84	<0.01	1.65	0.01	0.94	0.22
Combined arboreal locomotion variables	0.52	<0.01	0.76	<0.01	1.08	0.02	0.61	<0.01	1.55	<0.01	0.94	0.12
Positional variables (Hunt, 2016)												
sit	–	–	0.80	0.09	1.09	0.13	–	–	1.04	0.09	–	–
lie	–	–	0.08	0.08	1.04	0.22	–	–	0.83	0.36	–	–
q.stand	–	–	1.94	0.02	0.76	0.28	–	–	1.62	0.10	–	–
squat	–	–	0.65	0.22	0.85	0.22	–	–	1.07	0.09	–	–
cling	–	–	0.17	0.94	0.40	0.70	–	–	0.22	0.78	–	–
b.stand	–	–	0.92	0.07	1.68	0.12	–	–	0.21	0.88	–	–
armhang	–	–	1.50	<0.01	1.48	0.06	–	–	1.00	0.24	–	–
ips.suspend	–	–	1.30	0.02	1.58	0.12	–	–	none	none	–	–
q.walk	–	–	2.14	<0.01	1.09	0.05	–	–	0.85	0.25	–	–
vert.climb	–	–	0.49	0.32	0.74	0.26	–	–	0.59	0.33	–	–
leap	–	–	1.66	<0.01	1.63	0.07	–	–	1.65	0.03	–	–
q.run	–	–	0.18	0.88	1.20	0.11	–	–	0.22	0.76	–	–
b.walk	–	–	0.64	0.19	0.48	0.48	–	–	none	none	–	–
brachiate	–	–	1.38	<0.01	1.64	0.07	–	–	1.01	0.24	–	–
clamber	–	–	0.93	0.07	1.12	0.12	–	–	none	none	–	–
suspensory	–	–	0.36	0.68	0.46	0.65	–	–	0.22	0.75	–	–
transfer	–	–	0.27	0.72	0.34	0.79	–	–	none	none	–	–
sway	–	–	0.81	0.11	0.97	0.12	–	–	none	none	–	–
Combined positional variables (Hunt, 2016)	–	–	1.06	<0.01	1.16	0.01	–	–	0.92	0.02	–	–

Abbreviations: QTW = quadrupedal/tripedal walking; VCD = vertical climbing and descent; BW = bipedal walking; OCT = orthograde clamber and transfer; BFS = brachiation and forelimb swing; DL = drop and leap; q.stand = quadrupedal standing; b.stand = bipedal standing; arm.hang = arm hanging; ips.susp = ipsilateral suspension; q.walk = quadrupedal walking; vert.climb = vertical climbing; q.run = quadrupedal running; b.walk = bipedal walking.

[a] *Homo* was included only in one test (Anthropoids) to isolate the effect of the genus on subsequent results. $K < 1$ indicates less similarity, and a $K > 1$ indicates greater similarity, among closely related taxa than expected under Brownian motion. For Hunt (2016) data, we are unable to calculate K statistics for overall monkeys, ceboids and anthropoids (inclusive of *H. sapiens*) due to a lack of positional behavioral data for Ceboidea and *H. sapiens*, respectively. Some locomotor behaviors labeled 'none' were observed at 0 frequency for cercopithecoids.

[b] Definition of the groups: Anthropoids = total sample; Anthropoids (non-*Homo*) = all anthropoids excluding *Homo*; Hominoids (non-*Homo*) = all hominoids excluding *Homo*; Monkeys = all monkeys; Cercopithecoids = all Cercopithecoidea; Ceboids = all Ceboidea.

convergence on high intermembral and brachial proportions in the frequently suspensory and arboreal Asian apes in our sample. In the PCA of traditional indices, *Presbytis* is further removed from other cercopithecoids and all other monkeys than it is in the Mosimann PCA. This is because of *Presbytis*' unique combination of some of the lowest inter-limb proportions (i.e., relatively shortest forelimbs) and the highest brachial proportions (i.e., relatively longest forearms) in the sample, which is less apparent in PCAs based on size-adjusted long bone lengths.

4.2. How strong is the relationship between limb proportions and behavior?

Limb proportions and positional behaviors are strongly correlated across anthropoid taxa. Homoplasy between the Atelidae and *Pan*, and between the Hylobatidae and *Pongo*, underscores the strength of the selective pressures of forelimb-dominated, arboreal locomotive lifestyles, including orthograde clambering or suspensory behaviors, on larger-bodied primates (Pontzer and Wrangham, 2004; but see Cartmill, 1974; Fleagle, 1977; Jungers, 1985).

While limb proportions generally match contemporary behavioral repertoires, there are some exceptions. Several studies have reported that *Gorilla gorilla* and *Gorilla beringei* adults display different degrees of body size dimorphism and arboreality (Jungers et al., 1998; Jungers, 1985; Doran, 1996; Doran and McNeilage, 1998; Remis, 1999), but their limb length proportions are nearly identical (Taylor, 1997; see also shape differences of the scapulae: Remis, 1999; climatic adaptations reflected in limb proportions of *Gorilla* populations: Ruff et al., 2022). Although vertical climbing/descent and terrestrial quadrupedalism dominate the positional repertoires of adult *Pan*, their limb length proportions suggest a more careful, arboreal locomotor regime (Pontzer and Wrangham, 2004; see also Latimer, 1991; Hunt, 1991) that is driven by the safe harvesting of ripe fruit on terminal branches (Hunt, 2016), with evidence of age and sex differences in their arboreal activity patterns (Burgess, 2018; Doran, 1993b). These observations suggest that limb length proportions may be subject to phylogenetic inertia while maintaining behavioral efficiency at different life stages. However, the effects of well-recognized body shape and body mass-mediating environmental factors, like ambient temperature and oxygen pressure during development, have yet to be quantified for most primates (Allen, 1877; Bergmann, 1847; Ruff, 1994; Ruff et al., 2022).

The atelids *Alouatta* and *Lagothrix* have more *Pan*-like, elongated forelimb proportions than *Ateles*, but *Ateles* suspensory frequency is intermediate to *Pongo pygmaeus* and *Symphalangus*. Like other ceboids, *Alouatta* is most frequently an arboreal quadruped, while *Lagothrix* uses suspensory behaviors with a frequency on par with *Pan,* engages in relatively more orthograde climbing than any extant ape, and uses a degree of dropping and leaping consistent with New World monkeys. Napier and Napier (1967) regard *Presbytis* as a semi-brachiator, and observations in the wild report a small amount of forelimb-dominated behavior (Fleagle, 1977; Hunt, 2016; but see Byron et al., 2017). However, apart from its brachial index being in the range of the Hylobatidae (which is usually interpreted as indicative of forelimb-dominated arboreal proclivities: Schultz, 1930; Zapfe, 1961; Churchill et al., 2013), the limb proportions of *Presbytis* are similar to those of other monkeys and differ from those of other taxa with forelimb-dominated locomotor behaviors. The unique pattern of limb proportions in *Presbytis* may represent a compromise between an increased need to engage in some degree of suspensory behaviors, while phylogenetic inertia maintains efficient and safe leaping and dropping behaviors. Subsequent investigations focusing on monkeys will increase the sample to include other important taxa with diverse behavioral repertoires, like *Semnopithecus* and *Pygathrix*.

We found no sex differences in limb length proportions for any of the genera included in our comparative sample, and because our study only used adult specimens, any potential ontogenetic differences were unexplored. Sexual and ontogenetic effects on morphology and positional behaviors were not explored here because there were insufficient compatible sex-specific and developmental positional data for the majority of species in our dataset at the time this manuscript was prepared (but see Sarringhaus et al., 2014; Burgess, 2018 for studies of ontogenetic changes to behavior, long bone morphology, and long bone strength properties in *Pan*; and see Ruff et al., 2022 for *Gorilla*). We anticipate effects for sexually dimorphic species such as *Gorilla* and others that exhibit sex- and development-related behavioral biases linked to life history, and we recognize the potentially confounding effects of local environmental conditions on morphology and behavior (Jungers, 2009; Taylor, 1997; Young et al., 2010; Ruff et al., 2013; Druelle et al., 2016; Ruff et al., 2018; Young and Shapiro, 2018; Ruff et al., 2022).

4.3. How much of the observed variation in limb proportions and behavior is explained by phylogeny?

Phylogeny has a strong influence on most limb proportions and some behavioral repertoires in major taxonomic groups (e.g., anthropoids, hominoids, cercopithecoids, ceboids). In anthropoids, and in non-*Homo* anthropoids, VCD has a significant p-value ($p < 0.01$) with lower values of Blomberg's K ($K \leq 0.63$). This indicates that for VCD, selection is acting to keep some closely-related taxa more dissimilar than would be expected under Brownian motion. The exact implications of this phenomenon are unclear, but it is consistent with the well-recognized, fundamental importance of vertical climbing throughout primate evolution (Cartmill, 1974; Fleagle et al., 1981) and may reflect evolutionary histories characterized by new phenotypes arising due to adaptive radiations, resource competition, co-evolution with changing environments, or some combinations thereof in different lineages or at different times. Climbing has been implicated by some researchers as the immediate antecedent of both suspensory and bipedal locomotor types (Fleagle et al., 1981; Hunt et al., 1996; Lovejoy et al., 2009; Prost, 1980), and competition, especially with sympatric monkeys, has been cited as a major selective factor driving convergent increases in suspensory adaptations from climbing ancestors in hominoids and atelids (Hunt, 2016). The observed patterns support hypotheses implicating VCD as a fundamental locomotor behavior that is readily altered to accommodate phenotypic or environmental changes, and may give weight to hypotheses positing multiple reversals to more climbing-adapted phenotypes in hominins (e.g., OH 62 [Johanson et al., 1987]; BOU-VP-12/1 [Asfaw et al., 1999]; ARA-VP-6/500 [Lovejoy et al., 2009]).

Both apes and monkeys use high relative frequencies of vertical climbing and quadrupedal walking, but forelimb-dominated behaviors are relatively more important in hominoids than in non-atelid monkeys, whereas leaping behaviors are more common in monkeys. While selection has increased the ecological, morphological, and behavioral diversity of the few remaining extant ape genera, it seems to act on the many extant monkey genera to maintain relatively more generalized body plans and potentially less behavioral diversity than apes, despite the greater taxonomic and ecological diversity of extant monkeys. Although the origins of this hominoid diversity are unclear, the reduction and loss of the tail first evident in *Ekembo* (formerly *Proconsul*), and later acquisition of orthograde postures that appear with *Pierolapithecus*, *Hispanopithecus*, and *Oreopithecus* may have been parts of a suite of changes away from an ancestral arboreal above-branch quadrupedal body plan to one that is habitually orthograde (Almécija et al., 2021; Nakatsukasa, 2019; Ward, 1998). Young et al. (2010) indicated that evolutionary changes preceding the last common ancestors of extant hominoids also included reduced phenotypic integration (coordinated scaling) of the homologous forelimbs and hindlimbs via genetic decoupling. This was a major evolutionary step that has allowed their descendants to explore a greater range of behavioral repertoires and morphotypes than is evident in all extant hominoid forms. Whether a similar decoupling of limb element growth explains the unique limb length proportions evident in extant atelids and *Presbytis* remains to be determined.

5. Conclusion

Analyses of hominin associated skeletons indicate that fossil hominin limb proportions far exceeded intraspecific differences found in extant hominoids (Richmond et al., 2002), and that many fossil skeletons had combinations of limb proportions that are not represented among extant humans and other hominoids, such as purportedly mixed chimpanzee-like forelimb and human-like hindlimb proportions in the early hominin skeleton A.L. 288-1 (see Richmond et al., 2002), and human-like humerofemoral proportions paired with orangutan-like brachial proportions in the younger skeleton BOU-VP-12/1 (Asfaw et al., 1999; see also ARA-VP 6/500, White et al., 2009; LB 1, Brown et al., 2004; Sutikna et al., 2016; OH 62, Johanson et al., 1987; StW 573, Clarke and Tobias, 1995; Granger et al., 2023; Crompton et al., 2021; McHenry and Berger, 1998; Wood and Boyle, 2016). The seemingly incompatible limb segment proportions have inspired ongoing debate over hypotheses that explain hominin appendicular morphological evolution as punctuated by phenotypic reversions, and hypotheses positing hominin taxonomic diversity with instances of homoplasy. We interpret the remarkable variability of hominoid and fossil hominin limb proportions, combined with the evidence of phylogenetic inertia of limb length proportions demonstrated here and elsewhere (Young et al., 2010; Young and Shapiro, 2018), to support hypotheses of craniodental and postcranial homoplasy or parallelism in a taxonomically diverse hominin phylogeny. The strong association of appendicular shape and behavioral repertoires evident in extant taxa also supports hypotheses positing that hominin lineages with sufficiently dissimilar limb proportions experimented with unique behavioral patterns that included variable amounts and styles of bipedal locomotion. However, the links between limb length proportions and behavioral repertoires are imperfect and demonstrate once again that inter-limb length proportions, especially in the absence of intra-limb information, are not reliable indicators of adult behavior. The fragmented conditions of most extinct hominin skeletons have prevented direct measurements of most of their limb length proportions, which have been estimated by various quantitative and non-quantitative approaches with conflicting results (see radius length estimates for A.L.288-1, and questioned long bone maximum length estimates for OH 62 and BOU-VP-12/1 in Richmond et al., 2002). Future research will aim to develop quantitative and repeatable methods of reconstructing the limb length proportions of fossil hominins and use results to evaluate taxonomic hypotheses for fossil skeletons.

CRediT authorship contribution statement

Vance C.R. Powell: Conceptualization, Investigation, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing. **W. Andrew Barr:** Conceptualization, Writing – review & editing. **Ashley S. Hammond:** Conceptualization, Writing – review & editing. **Bernard A. Wood:** Conceptualization, Writing – review & editing.

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Supplementary Online Material

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